R94 | I/O PIPELINE RECOVERY

2026-10-22 | Accepted B startup restored | Not model-quality evidence

2026-10-22 | Accepted B startup throughput restored; not evidence of model-quality improvement.

Evidence and Startup Scope

The authoritative archive is startup_slow_io_attempt1/ARCHIVE.json under /public/home/shiys/0selection2/code/V4/runs/R94_pretrained_node_fusion. Main stopped only the very slow first R94B startup and the unstarted R94A queue. R92A/B and R93A remained healthy and untouched per the handoff; this report did not poll them.

The archived B log records 108 successful updates; actual executed updates are bounded at 108-125, not known exactly. Durable last.pt has 0 successful updates and an empty optimizer state. There was no VAL, TEST or LIB evaluation. The startup is discarded, not resumed. Original controller/worker SIGTERM failures remain in /public/home/shiys/0selection2/code/V4/runs/R94_launch/startup_slow_io_attempt1.

Archived steps 18/36/54/72 report 7.489/7.407/7.408/7.335 seconds/update; step 1 is 6.988 and step 108 is 7.647. The handoff reports an idle GPU with about 19 GB allocated. These are before-fix observations.

Bounded CPU Input Microbenchmark

Supplied Sol measurements used the job-1475-owned container, unused CPU 74, CPU input only, no GPU or labels. Each batch requested 1024 x 101 x 128 FP16 values: 26,476,544 bytes. Disjoint fixed-random TRAIN rows were read while other jobs ran: not paired, controlled tests or training-throughput measurements. This worker did not rerun them.

I/O policy	Record	Seconds	Physical read bytes	Phys./req.
NORMAL	1	12.626988	2,680,324,096	101.234x
NORMAL	2	6.585618	1,203,400,704	45.452x
MADV_RANDOM only	1	29.093819	18,288,640	0.691x
MADV_RANDOM only	2	16.189236	9,932,800	0.375x
RANDOM + span WILLNEED	1	0.102639	30,584,832	1.155x
RANDOM + span WILLNEED	2	0.112926	30,564,352	1.154x

Physical/requested = physical read bytes / 26,476,544. Ratios below one can reflect cache-served pages, not missing values. Exact times and bytes remain in io_readaround_evidence.json.

MADV_RANDOM alone was rejected: fewer bytes but slower than NORMAL in both records. Byte reduction alone is not a performance win. CPU probe latency is separate from the actual-run observation below.

Observed End-to-End Startup Recovery

Root io_runtime_recovery.json records accepted B at 270 successful updates, phase=training, all 18 tasks finite, checked at 2026-10-21T21:14:58.827089+00:00. Its complete contents are embedded in the published evidence JSON, so this report does not depend on that sidecar being published separately.

Observation	Original B startup	Accepted fresh B
Logged timing window, successful updates	54	270
Seconds per successful update	7.4084640651	0.4247711723
Feature-cache preparation, seconds	747.88	0.5115 (reused)

Operational throughput restored: observed old/new ratio 17.4411x, approximately 17.44x, from exact JSON times. Old54 predates the extra CPU I/O probe; new270 is a different window. Co-resident jobs were active: this is not a controlled benchmark, a precisely established causal factor, or model-quality evidence. Old cache-prep time is supplied handoff evidence; the reused time is in the runtime JSON. No cache regeneration occurred.

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Adopted I/O Policy and Scientific Boundary

The sidecar combines mapping-level MADV_RANDOM, descriptor-level POSIX_FADV_RANDOM, and asynchronous POSIX_FADV_WILLNEED for explicit requested-row spans before unchanged gather. It limits read-around while allowing queued parallel I/O, rather than random-only demand faults.

Feature content, sample ordering, label/padding rules, RNG setup, logical batch and loss/gradient definitions are unchanged. This does not assert identical floating-point GPU values. No native online encoder fallback is introduced. The approximately 41 GB TRAIN/VAL cache at /tmp/shiys_R94_B_node_features is preserved, not regenerated.

Main reports exact equality of scientific source hashes/metadata for model, loss, initialization, native encoder and sample order; launcher, inference and report sources also remain unchanged. Only sidecar I/O and publication files changed. This worker did not rehash sources, weights or arrays; queue/source freeze remain with main. Original protocol-lock SHA256: 54ea04fbdf06d6d63d38550b6f313e4b3b10648dc9f29f307064e245a6846d85.

Fresh Restart and Budget Boundary

Accepted A and B use fresh starts from the same original locked seed-2 initialization under the unchanged protocol. Neither resumes discarded trained state: this is resource correction, not numerical-failure recovery, scientific reselection or automatic retry. Old A had not started; the actual-GPU recovery observation above is B only.

B's archived configured budget is 57,600 successful updates. The discarded 108-125 updates are separate recovery overhead (0.1875%-0.2170% of that B budget), not accepted progress, a checkpoint candidate, or a deduction from the fresh accepted budget. At logical batch 1,024, this bounds discarded sample exposures at 110,592-128,000. No accepted-run budget or selection rule is changed by this report.

CPU/Metadata Checks and GPU Numerical Agreement

The handoff reports byte-for-byte CPU input parity, row order and file-descriptor cleanup checks passing. Full-suite counts remain owned by main in root cpu_checks.json; they were not read, inferred or awaited for this update. CPU input parity does not imply bitwise GPU arithmetic.

The initial trainable-parameter SHA256 is identical: 9db3df55a91ca0f331880cfcc4688ffab82b83bb66e80f9cebe9d6978e903cb4. At common logged steps 1/18/36/54/72/90/108, maximum absolute loss difference is 3.1661518196e-7 (<=3.17e-7), and grad-norm difference is 1.0379652182e-4 (<=1.04e-4). These are close, not bitwise-equal GPU results; all objective-part differences are retained in the embedded runtime JSON.

No VAL or TEST was read/used for the fix. Startup throughput is restored at the recorded observation, not a model-quality comparison, completed-model result, VAL/TEST/LIB result or native-online performance conclusion.

Report-Only Scope and Provenance

Only this Markdown, the two-page IO_PIPELINE_RECOVERY.pdf and io_readaround_evidence.json are written. Commands use easynco, empty CUDA_VISIBLE_DEVICES, and threads <=2. No frozen-runtime/model edits, GPU, SSH, raw arrays, weights, forward passes, profiling, training, launch, commit or subagents; no other changes reverted.

Archive/log facts and the small runtime JSON were read locally. CPU probes, checks, peer health, source equality and cache provenance remain main-controller evidence, not independently re-executed here. Actual GPU observations come from the recorded accepted run, not a new report-side run. PDF uses the existing ADMIN CJK font and Agg/embedded-font setup.